
Bayesian Networks: Sampling

CS5491: Artificial Intelligence
ZHICHAO LU

Content Credits: [Prof. Wei's CS4486 Course](#)
and [Prof. Boddeti's AI Course](#)



TODAY

Prior Sampling

Rejection Sampling

Likelihood Weighting

Gibbs Sampling



RECAP: BAYES NET REPRESENTATION

A directed, acyclic graph, one node per random variable

A conditional probability table (CPT) for each node

- A collection of distributions over $\mathbf{X}$, one for each combination of parents values

$$P(X|a_1, \dots, a_n)$$

Bayes nets implicitly encode joint distributions

- As a product of local conditional distributions
- To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(x_i))$$



RECAP: VARIABLE ELIMINATION

Interleave joining and marginalizing

d^k entries computed for a factor over k variables with domain sizes d

Ordering of elimination of hidden variables can affect size of factors generated

Worst case: running time exponential in the size of the Bayes net



APPROXIMATE INFERENCE: SAMPLING



SAMPLING

Why sample?



SAMPLING

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- Learning: get samples from a distribution you don't know
- Inference: getting a sample is faster than computing the right answer (e.g. with variable elimination)



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Sampling is a lot like repeated simulation

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Basic idea

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Basic idea

- Draw N samples from a sampling distribution S
- Compute an approximate posterior probability
- Show this converges to the true probability P

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SAMPLING

Sampling from given distribution



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- Step 1: Get sample u from uniform distribution over $[0, 1)$
 - e.g. *random()* in python



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Example:

C	$P(C)$
<i>red</i>	0.6
<i>green</i>	0.1
<i>blue</i>	0.3

$$0 \leq u < 0.6, \rightarrow C = \textit{red}$$

$$0.6 \leq u < 0.7, \rightarrow C = \textit{green}$$

$$0.7 \leq u < 1.0, \rightarrow C = \textit{blue}$$



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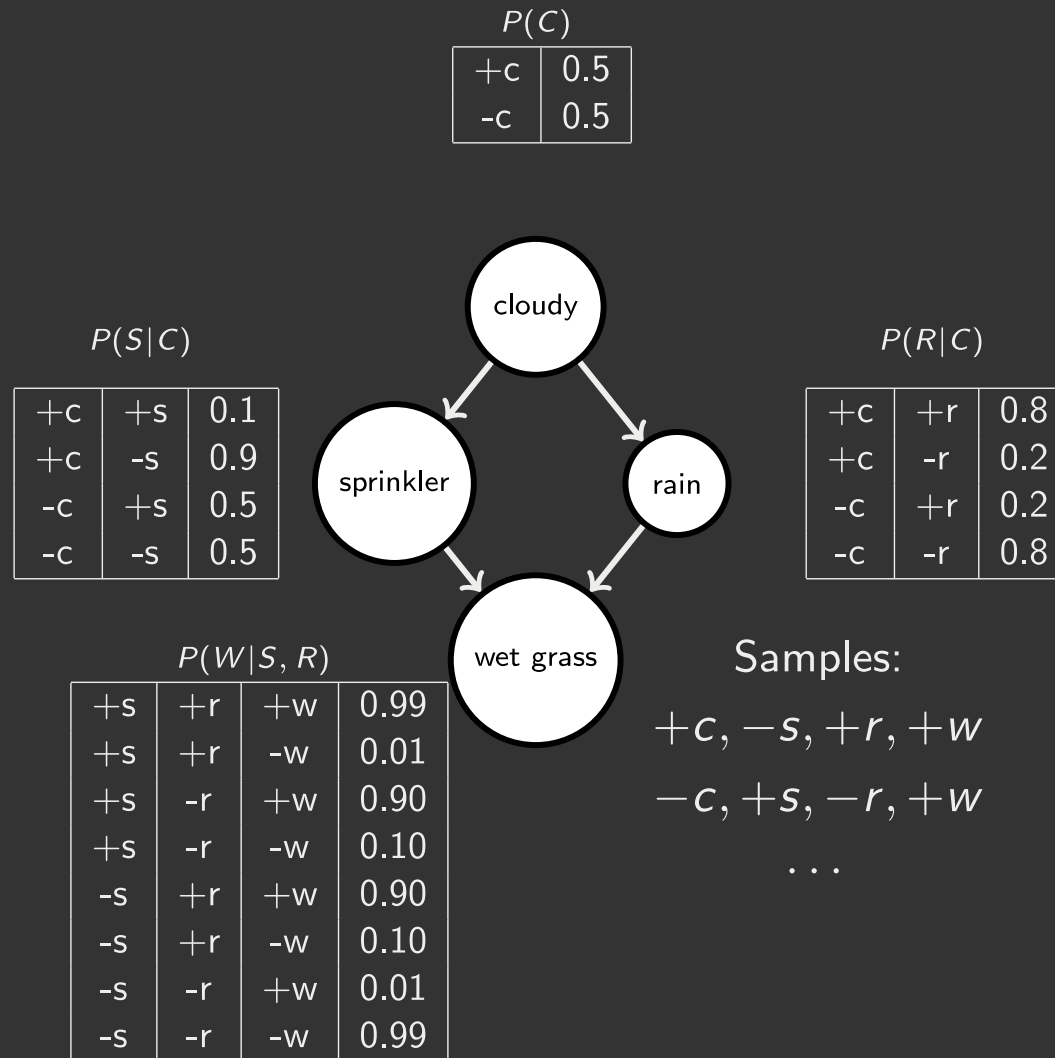
- If *random()* returns $u = 0.83$, then our sample is $C = \textit{blue}$.



PRIOR SAMPLING



PRIOR SAMPLING



PRIOR SAMPLING

Prior Sampling:

```
Function prior-sampling(bayes net)  
  for  $i = 1, 2, \dots, n$  do  
    | Sample  $x_i$  from  $P(X_i | Parents(X_i))$  ;  
    | return  $(x_1, x_2, \dots, x_n)$ ;  
  end
```

PRIOR SAMPLING

This process generates samples with probability:

$$S_{PS}(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(X_i)) = P(x_1, \dots, x_n)$$

i.e., the BN's joint probability.



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Then,

$$\begin{aligned} \lim_{N \rightarrow \infty} \hat{P}(x_1, \dots, x_n) &= \lim_{N \rightarrow \infty} \frac{N_{PS}(x_1, \dots, x_n)}{N} \\ &= S_{PS}(x_1, \dots, x_n) \\ &= P(x_1, \dots, x_n) \end{aligned}$$

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i.e., the sampling procedure is **consistent**



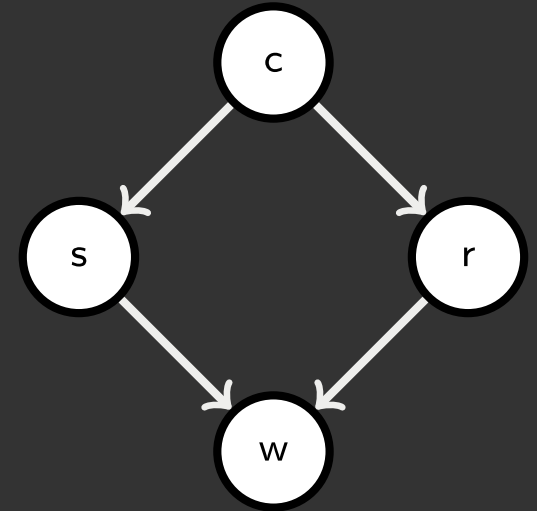
EXAMPLE: PRIOR SAMPLING

We will get a bunch of samples from the BN:

- $+c, -s, +r, +w$
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If we want to know $P(W)$

- We have counts $\langle +w : 4, -w : 1 \rangle$
- Normalize to get $P(W) = \langle +w : 0.8, -w : 0.2 \rangle$
- This will get closer to the true distribution with more samples
- Can estimate anything else, too



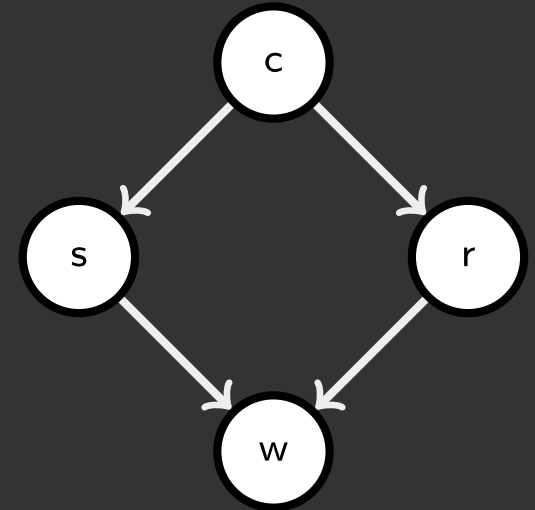
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- What about $P(C | +w)$? $P(C | +r, +w)$? $P(C | -r, -w)$?



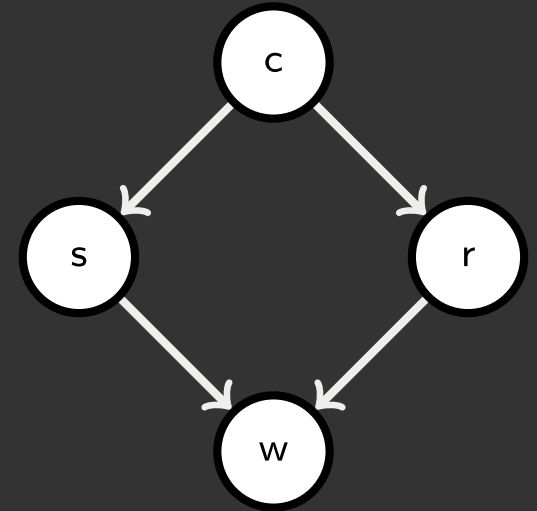
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- Can estimate anything else, too
- What about $P(C | +w)$? $P(C | +r, +w)$? $P(C | -r, -w)$?
- Fast: can use fewer samples if less time (what is the drawback?)



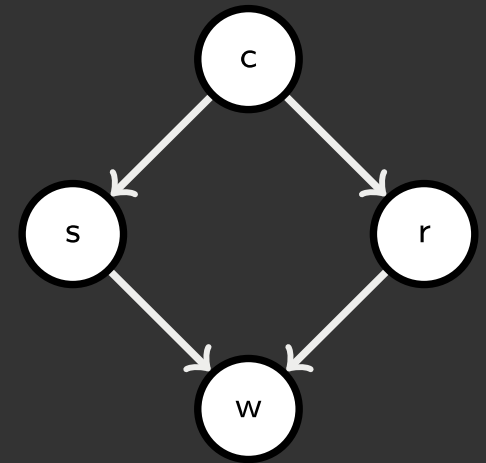
REJECTION SAMPLING



REJECTION SAMPLING

Let us say we want $P(\mathcal{C})$

- No point keeping all samples around
- Just tally counts of $\mathcal{C}$ as we go



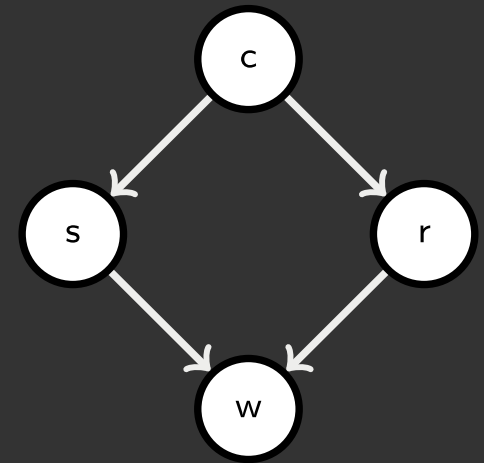
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Let us say we want $P(C|+s)$



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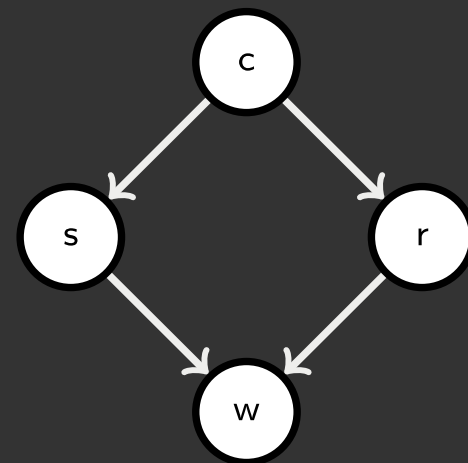
REJECTION SAMPLING

Let us say we want $P(C)$

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Let us say we want $P(C | +s)$

- Same thing: tally C outcomes, but ignore (reject) samples which do not have $S = +s$
- This is called **rejection sampling**
- It is also consistent for conditional probabilities (i.e., correct in the limit)



$+c, -s, +r, +w$
 $+c, +s, +r, +w$
 $-c, +s, +r, -w$
 $+c, -s, +r, +w$
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REJECTION SAMPLING

Rejection Sampling:

```
Function rejection-sampling(bayes-net,  
    evidence-instantiation)  
    for  $i = 1, 2, \dots, n$  do  
        Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        if  $x_i$  not consistent with evidence then  
            | Reject: return – no sample is generated in this cycle  
        end  
        return  $(x_1, x_2, \dots, x_n)$ ;  
    end
```

LIKELIHOOD WEIGHTING



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Problem with rejection sampling:



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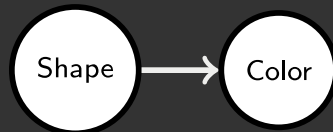
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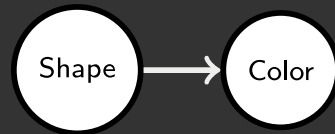
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- Consider $P(\textit{Shape}|\textit{blue})$



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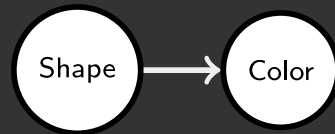
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Idea: fix evidence variables and sample the rest

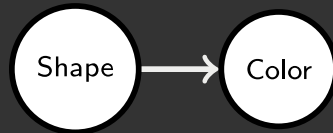


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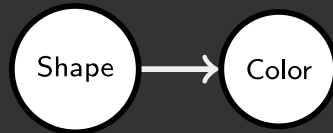
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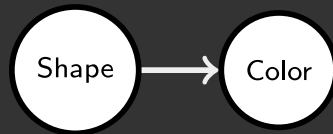
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- Problem: sample distribution not consistent
- Solution: weight by probability of evidence given parents

LIKELIHOOD WEIGHTING

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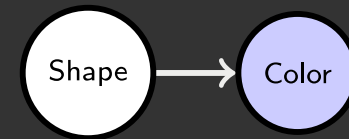
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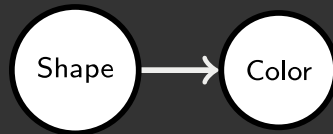
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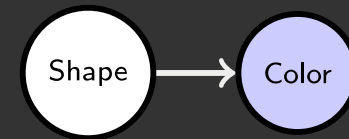
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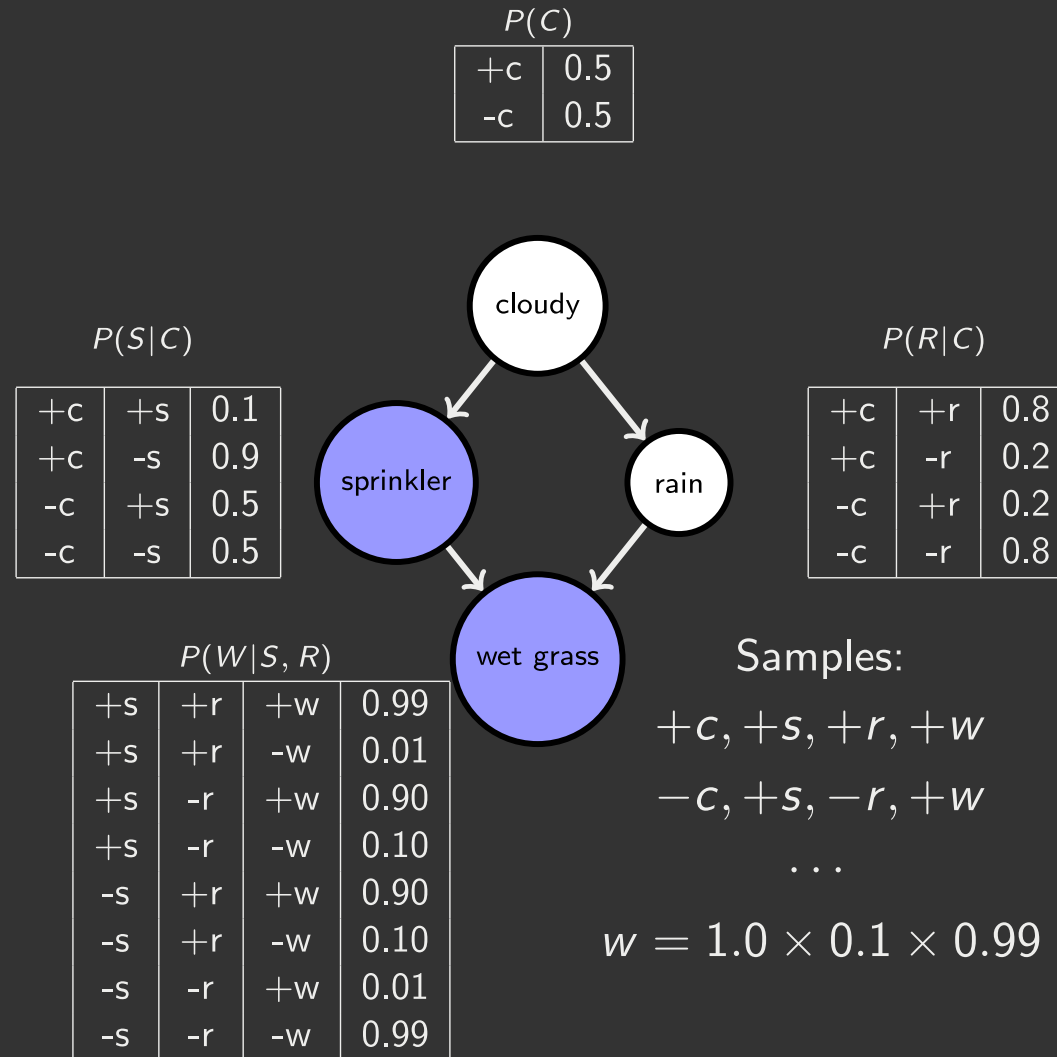
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LIKELIHOOD WEIGHTING: EXAMPLE



LIKELIHOOD WEIGHTING: ALGORITHM

Likelihood Weighting:

```
Function likelihood-weighting(bayes-net,  
    evidence-instantiation)  
     $w = 1.0$  ;  
    for  $i = 1, 2, \dots, n$  do  
        Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        if  $x_i$  is an evidence variable then  
             $X_i = \text{observation } x_i \text{ for } X_i$  ;  
            Set  $w = w * P(x_i | \text{Parents}(X_i))$  ;  
        end  
        else  
            Sample  $x_i$  from  $P(X_i | \text{Parents}(X_i))$  ;  
        end  
    return  $(x_1, x_2, \dots, x_n), w$  ;  
end
```

LIKELIHOOD WEIGHTING

Sampling distribution if $\mathbf{z}$ sampled and $\mathbf{e}$ fixed evidence

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$



LIKELIHOOD WEIGHTING

Sampling distribution if $\mathbf{z}$ sampled and $\mathbf{e}$ fixed evidence

$$S_{WS}(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^l P(z_i | \text{Parents}(Z_i))$$

Now, samples have weights

$$w(\mathbf{z}, \mathbf{e}) = \prod_{i=1}^m P(e_i | \text{Parents}(E_i))$$

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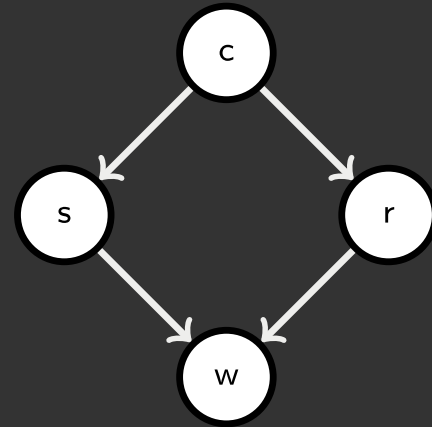
Together, weighted sampling distribution is consistent

$$\begin{aligned} S_{WS}(z, e) \cdot w(z, e) &= \prod_{i=1}^l P(z_i | Parents(z_i)) \prod_{i=1}^l P(e_i | Parents(e_i)) \\ &= P(\mathbf{z}, \mathbf{e}) \end{aligned}$$



LIKELIHOOD WEIGHTING

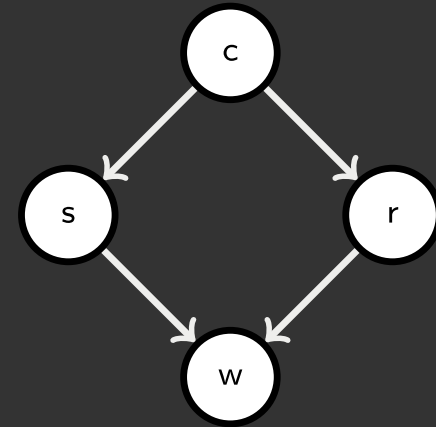
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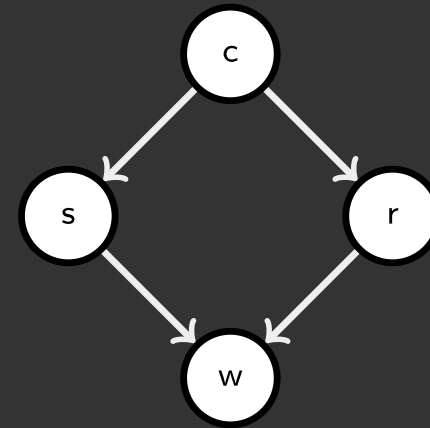
- We have taken evidence into account as we generate the sample
- E.g. here, W 's value will get picked based on the evidence values of S , R
- More of our samples will reflect the state of the world suggested by the evidence



LIKELIHOOD WEIGHTING

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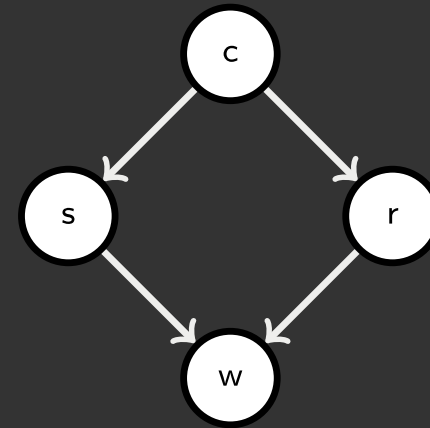


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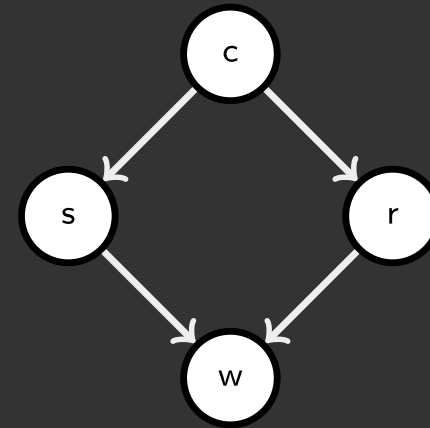
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- Evidence influences the choice of downstream variables, but not upstream ones (C is not more likely to get a value matching the evidence)

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We would like to consider evidence when we sample every variable (leads to Gibbs sampling)

GIBBS SAMPLING



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Procedure: keep track of a full instantiation $x_1, x_2, \dots, x_n$. Start with an arbitrary instantiation consistent with the evidence. Sample one variable at a time, conditioned on all the rest, but keep evidence fixed. Keep repeating this for a long time.



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Property: in the limit of repeating this infinitely many times the resulting samples come from the correct distribution (i.e. conditioned on evidence) variable.



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Rationale: both upstream and downstream variables condition on evidence

In contrast: likelihood weighting only conditions on upstream evidence, and hence weights obtained in likelihood weighting can sometimes be very small. Sum of weights over all samples is indicative of how many ``effective" samples were obtained, so we want high weight.



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

Step 1: Fix evidence



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

Step 1: Fix evidence

- $R = +r$



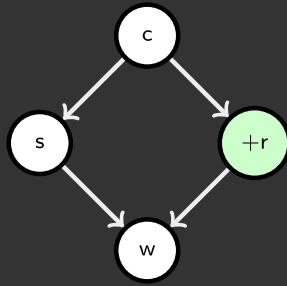
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Step 2: Initialize other variables

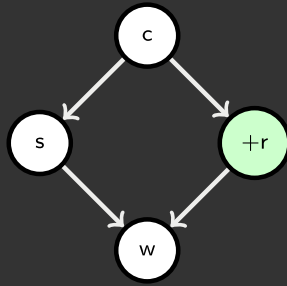
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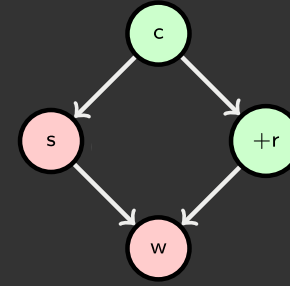
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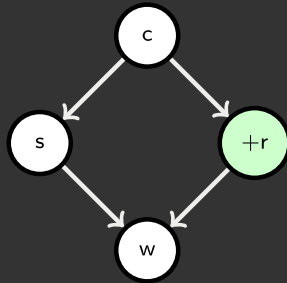
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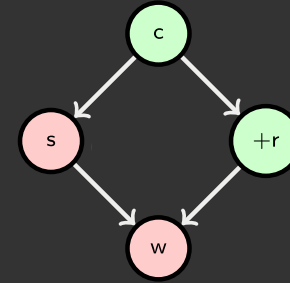
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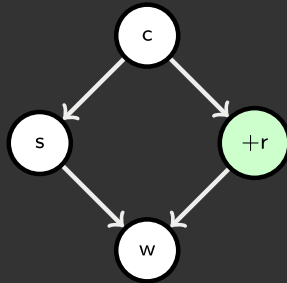


Step 3: Repeat

GIBBS SAMPLING EXAMPLE: $P(S|+R)$

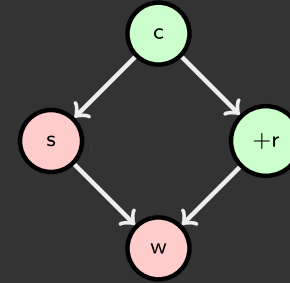
Step 1: Fix evidence

- $R = +r$



Step 2: Initialize other variables

- randomly



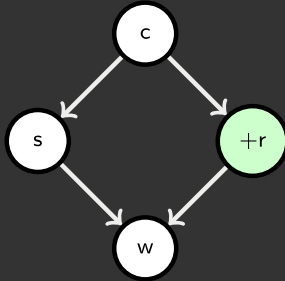
Step 3: Repeat

- Choose a non-evidence variable X

GIBBS SAMPLING EXAMPLE: $P(S|+R)$

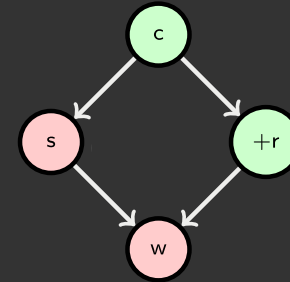
Step 1: Fix evidence

- $R = +r$



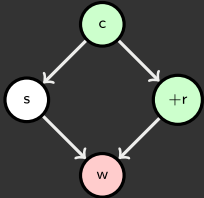
Step 2: Initialize other variables

- randomly



Step 3: Repeat

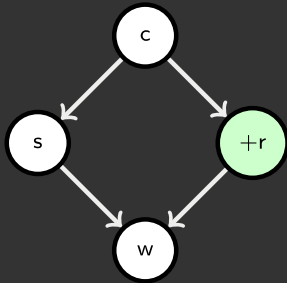
- Choose a non-evidence variable X



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

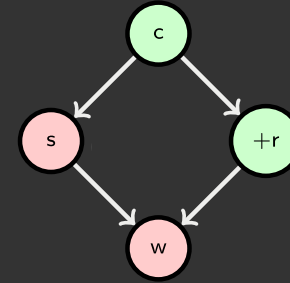
Step 1: Fix evidence

- $R = +r$



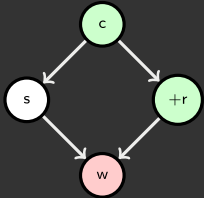
Step 2: Initialize other variables

- randomly



Step 3: Repeat

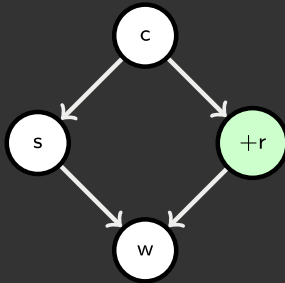
- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

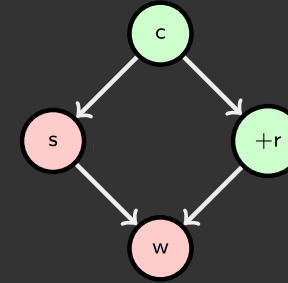
Step 1: Fix evidence

- $R = +r$



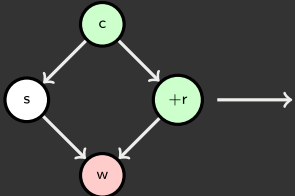
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



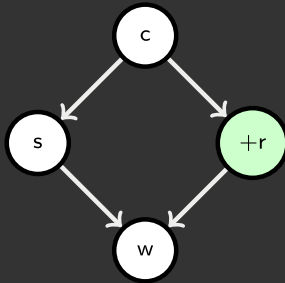
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

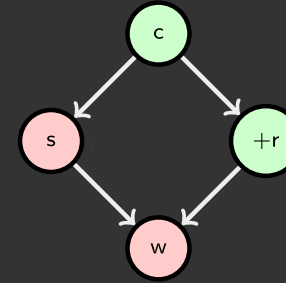
Step 1: Fix evidence

- $R = +r$



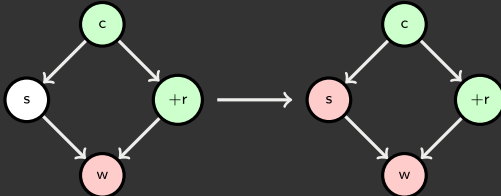
Step 2: Initialize other variables

- randomly



Step 3: Repeat

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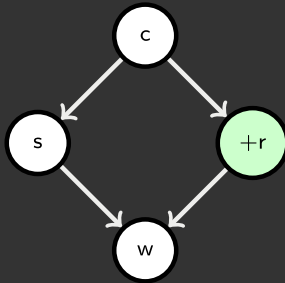
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

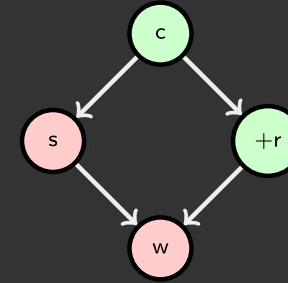
Step 1: Fix evidence

- $R = +r$



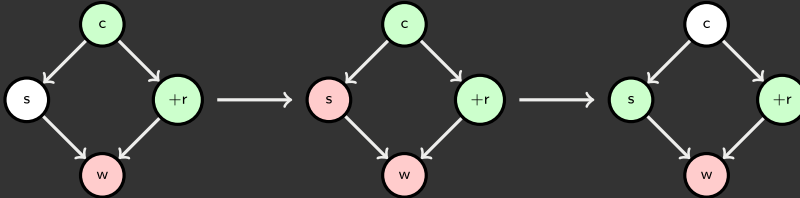
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



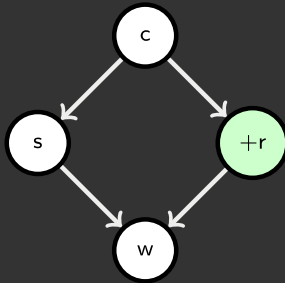
Sample from $P(S|+c, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

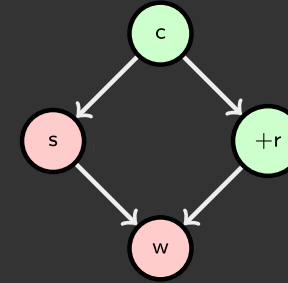
Step 1: Fix evidence

- $R = +r$



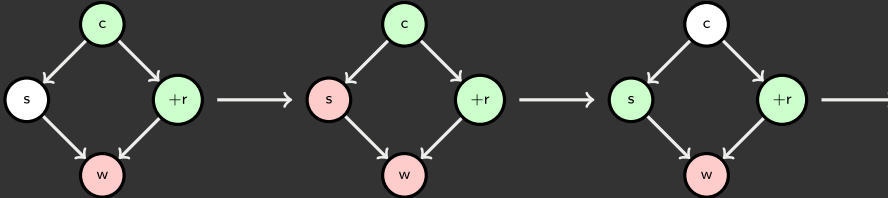
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|c, -w, +r)$

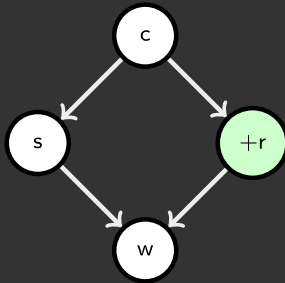
Sample from $P(C|s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

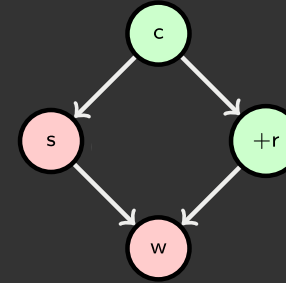
Step 1: Fix evidence

- $R = +r$



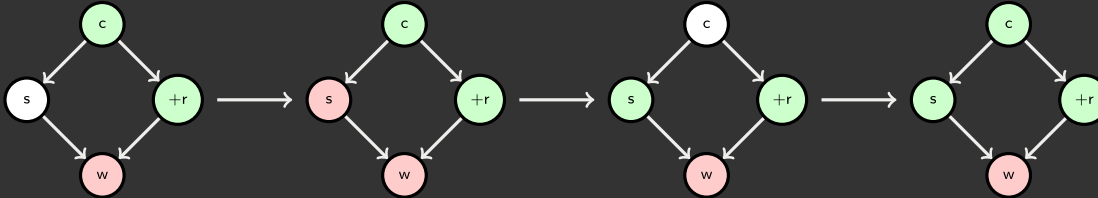
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|c, -w, +r)$

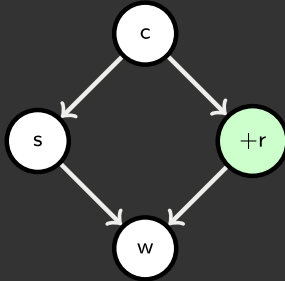
Sample from $P(C|s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

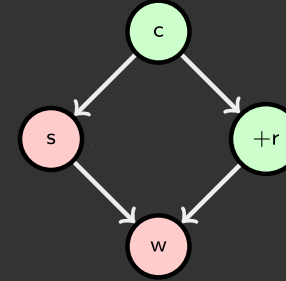
Step 1: Fix evidence

- $R = +r$



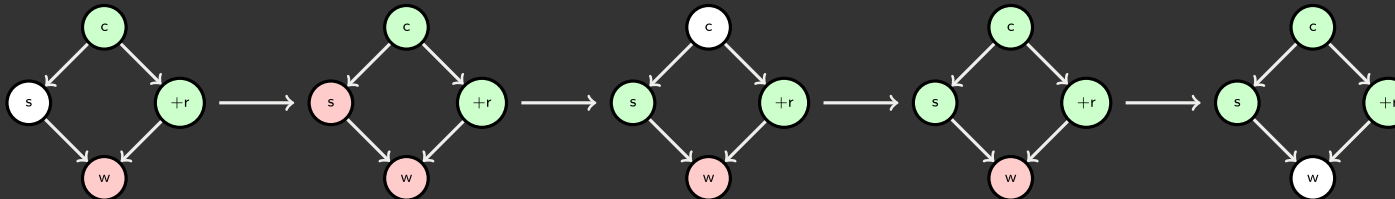
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Sample from $P(S|+c, -w, +r)$

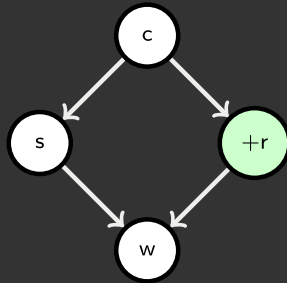
Sample from $P(C|+s, -w, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

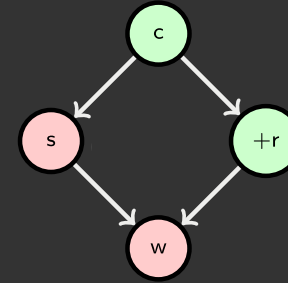
Step 1: Fix evidence

- $R = +r$



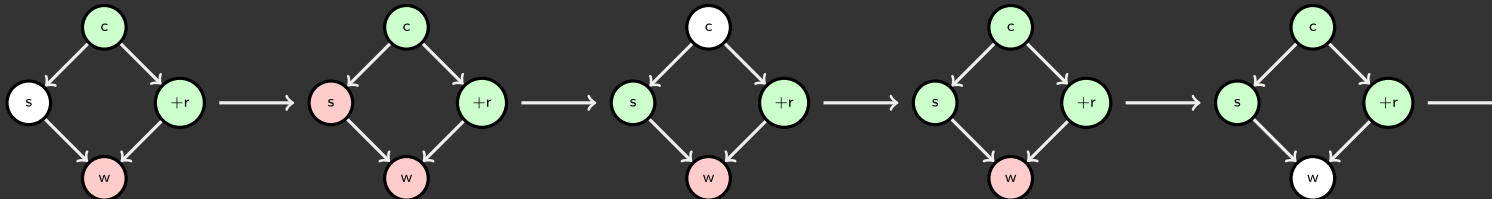
Step 2: Initialize other variables

- randomly



Step 3: Repeat

- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

Sample from $P(C|+s, -w, +r)$

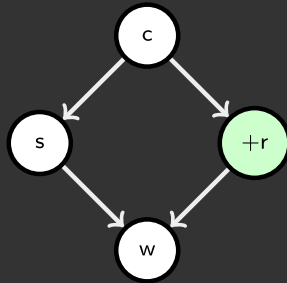
Sample from $P(W|+s, +c, +r)$



GIBBS SAMPLING EXAMPLE: $P(S|+R)$

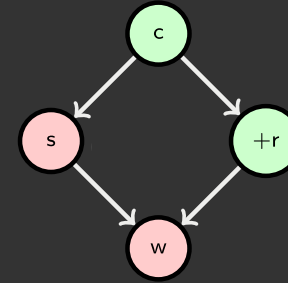
Step 1: Fix evidence

- $R = +r$



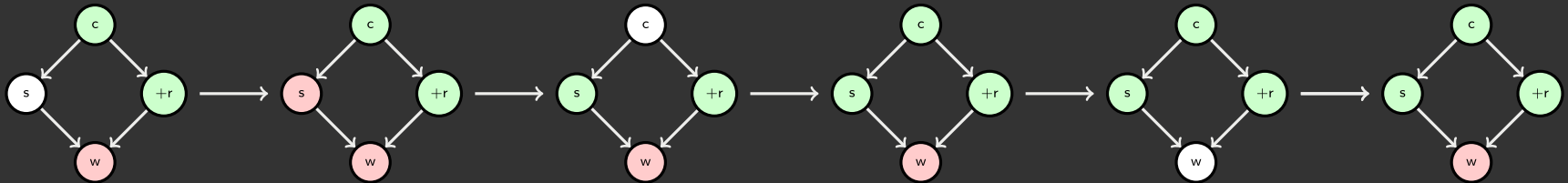
Step 2: Initialize other variables

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- Choose a non-evidence variable X
- Resample X from $P(X|\text{all other variables})$



Sample from $P(S|+c, -w, +r)$

Sample from $P(C|+s, -w, +r)$

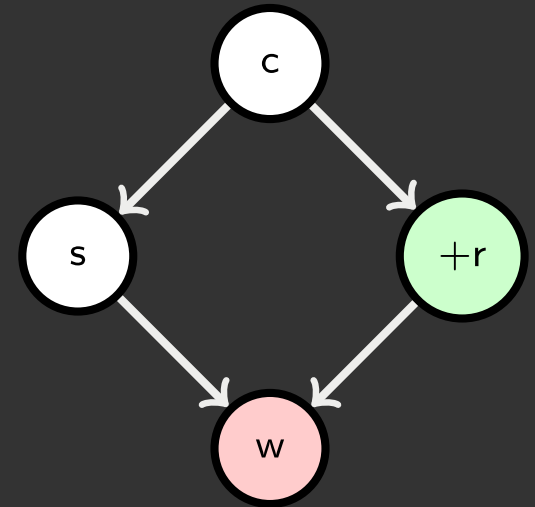
Sample from $P(W|+s, +c, +r)$



EFFICIENT RESAMPLING OF ONE VARIABLE

Sample from $P(S \mid +c, +r, -w)$

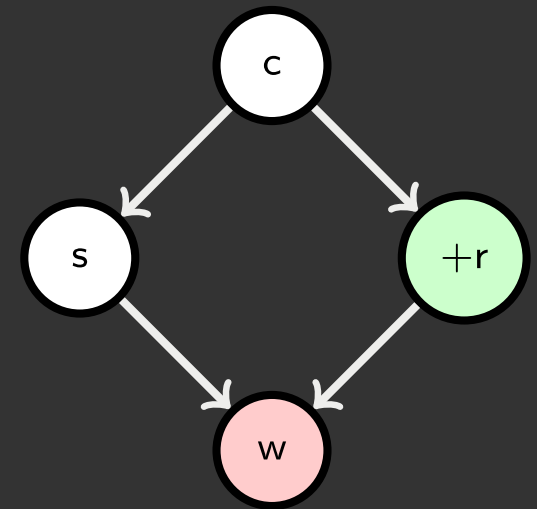
$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(S \mid +c)P(-w \mid s, +r)} \end{aligned}$$



EFFICIENT RESAMPLING OF ONE VARIABLE

Sample from $P(S \mid +c, +r, -w)$

$$\begin{aligned} P(S \mid +c, +r, -w) &= \frac{P(S, +c, +r, -w)}{P(+c, +r, -w)} \\ &= \frac{P(S, +c, +r, -w)}{\sum_s P(s, +c, +r, -w)} \\ &= \frac{P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid S, +r)}{\sum_s P(+c)(S \mid +c)P(+r \mid +c)P(-w \mid s, +r)} \\ &= \frac{P(S \mid +c)P(-w \mid S, +r)}{\sum_s P(S \mid +c)P(-w \mid s, +r)} \end{aligned}$$



Many things cancel out — only CPTs with S remain

More generally: only CPTs that have resampled variable need to be considered, and joined together

SUMMARY: BAYES NET SAMPLING

Prior Sampling $P(Q)$

Rejection Sampling $P(Q|e)$

Likelihood Weighting $P(Q|e)$

Gibbs Sampling $P(Q|e)$



BEYOND GIBBS SAMPLING

Gibbs sampling produces sample from the query distribution $P(Q|e)$ in limit of re-sampling infinitely often



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Gibbs sampling is a special case of more general methods called Markov chain Monte Carlo (MCMC) methods

- Metropolis–Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis–Hastings)



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- Metropolis–Hastings is one of the more famous MCMC methods (in fact, Gibbs sampling is a special case of Metropolis–Hastings)

You may read about Monte Carlo methods — they are just sampling



